MODULE 02 · DOMAIN 1 - DATA PREPARATION FOR ML AND AI

Storage and ingestion for ML

Where training data lives, and how it reaches the training job.

MODULE 02 of 43 DOMAIN 1 - Data Prep WEIGHT 28%

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A training job that is **waiting for data** is still being billed for its GPUs.

How data reaches a training container

01

Source

S3, EFS or FSx for Lustre. Only these three.

02

Mode

Download it, stream it, or mount it.

03

Container

The script sees a normal file path either way.

04

Output

Model artefacts and checkpoints go back to S3.

ON THE EXAM

Source and mode are two separate decisions. Questions often change one and leave the other alone.

Why Amazon S3 is where ML data lives

- **Effectively unlimited**, and eleven nines of durability. Datasets grow without planning.
- **Decoupled from compute.** Terminate the cluster, keep the data.
- Every AWS analytics and ML service reads from it. Glue, Athena, EMR, SageMaker AI.
- **Lifecycle policies** move ageing data to cheaper classes with no code.
- It is the assumed answer. Choosing something else needs a reason from the scenario.

> ON THE EXAM
>
> If a question does not give you a reason to leave S3, do not leave S3.

Making the initial storage decision

- **Cost.** How much, how long, and how often will it actually be read?
- **Performance.** Is anything waiting on this data, and how fast must it arrive?
- **Structure.** Tabular, images, free text, or a mixture? That drives format and layout.
- Data gravity is real. Whatever generates the data usually decides where it first lands.
- The guide lists this as a skill in its own right: initial storage decisions on cost, performance and structure.

> ON THE EXAM
>
> Moving data later is cheap for gigabytes and painful for petabytes. The first decision is the one that sticks.

The S3 classes that come up in ML questions

CLASS	REACH FOR IT WHEN
S3 Standard	Active training data. The default, and usually correct
S3 Intelligent-Tiering	Access pattern is unknown or irregular. Automatic, small monitoring fee
S3 Standard-IA	Older training sets kept for reproducibility. Retrieval costs money
S3 Express One Zone	Very high request rates, single AZ, latency-sensitive access
S3 Glacier Flexible Retrieval	Archived raw data. Minutes to hours to get it back
S3 Glacier Deep Archive	Compliance retention. Hours to restore. Cheapest storage

ON THE EXAM

Archive classes are for data you are keeping, not data you are training on. Retrieval time makes them useless mid-job.

S3 performance, and one myth to drop

- S3 scales to **5,500 GET** and **3,500 PUT** requests per second **per prefix**.
- More prefixes means more parallel throughput. Spread reads across many prefixes.
- **The old advice to randomise key prefixes is obsolete.** S3 partitions automatically now.
- **Multipart upload** for large objects: parallel parts, and a failed part retries alone.
- Byte-range fetches let readers pull parts of one large object in parallel.

> ON THE EXAM
>
> Date-based or class-based prefixes are fine and readable. You do not need hashed keys any more.

S3 Express One Zone, and its trap

- Uses **directory buckets**, not general purpose buckets. Single Availability Zone.
- Built for **very high request rates** and single-digit millisecond access.
- One AZ means **no cross-AZ resilience**. Lose the AZ, lose the data.
- **SSE-KMS is not supported** for SageMaker AI output data in directory buckets.
- Higher storage price per GB, lower request price. It pays off only at high request volume.

> ON THE EXAM
>
> A compliance requirement for customer-managed KMS keys on training output rules out directory buckets outright.

Where the data lives and **how it arrives** are different questions with different answers.

Three ways data reaches the container

MODE	WHAT ACTUALLY HAPPENS
File	Downloads the entire dataset to local disk before your script starts. The default
FastFile	Streams from S3 on demand. Script starts immediately. POSIX interface
Pipe	Streams from S3 into FIFO named pipes. Reduces the EBS volume you need
EFS	Mounts an existing file system. Data must already be there. Needs a VPC
FSx for Lustre	Mounts a high-throughput file system. Needs a VPC. Mount time is size-independent

ON THE EXAM

File, FastFile and Pipe apply to S3 only. EFS and FSx are mounted, so the mode question does not arise.

File mode or FastFile mode?

FILE MODE, WHEN

Small, or read many times

- The dataset is small enough to download fast.
- Many small files, read repeatedly per epoch.
- You want the simplest possible setup.
- Local disk random reads matter to you.

FASTFILE MODE, WHEN

Large files, read once

- Files are over 50 MB, ideally over 150 MB.
- Startup delay is the problem you are solving.
- Reads are mostly sequential.
- You want streaming without a file system.

THE TIEBREAKER

AWS guidance: for large datasets made of large files, try fast file mode first. It needs no file system and no VPC.

Pipe mode, and why you see it less now

- Streams S3 data into **FIFO named pipes** at high concurrency.
- Cuts the EBS volume you must provision, since nothing lands on disk.
- **Sequential only.** No random access, no shuffling within the stream.
- Needs a supported format such as RecordIO-protobuf, or an augmented manifest.
- FastFile covers most of what Pipe was built for, with far less friction.

> ON THE EXAM
>
> If a question stresses reducing storage attached to the training instance, Pipe mode is the intended answer.

FSx for Lustre earns its keep on repetition

- Hundreds of GB per second of throughput, millions of IOPS. Built for HPC.
- **Links to an S3 bucket.** Objects copy across on first access, then stay cached.
- That caching is the point: **run after run over the same data** pays no download cost.
- Mount time does not depend on dataset size. A huge dataset mounts as fast as a small one.
- Handles **many small files** well, which is where File mode struggles most.

> ON THE EXAM
>
> The signal is iteration. One job over a dataset does not justify FSx. Fifty jobs over the same dataset does.

What FSx for Lustre costs you

- **Requires a VPC connection** from the training job. That is real DevOps setup.
- **Single Availability Zone**, to avoid data transfer charges. You pin the subnet to that AZ.
- You provision and pay for the file system **continuously**, not per training job.
- Minimum size is **1.2 TB**, whether your dataset needs it or not.
- Scratch 2 gives 200 MB/s per TB baseline, bursting to 1300 MB/s per TB.

> ON THE EXAM
>
> A file system billing around the clock for one monthly training run is a cost question waiting to happen.

FSx for Lustre or Amazon EFS?

FSX FOR LUSTRE, WHEN

Throughput is the problem

- Training is starved waiting for data.
- Large dataset, many repeated jobs.
- You want the S3 link and its caching.
- You accept single-AZ and 1.2 TB minimum.

AMAZON EFS, WHEN

The data is already there

- Preprocessing or labelling already writes to EFS.
- Several teams share one evolving dataset.
- You want elastic capacity, no provisioning.
- Multi-AZ durability matters more than raw speed.

THE TIEBREAKER

EFS is rarely chosen for speed. It is chosen because the data is already sitting there and moving it would be silly.

FSx for NetApp ONTAP, and why the guide names it

- A managed **NetApp ONTAP** file system, speaking NFS, SMB and iSCSI.
- Named explicitly in the exam guide as a core AWS data source.
- The case is **migration**: an enterprise already runs ONTAP on premises.
- It is not a performance answer for training. FSx for Lustre is the performance answer.
- Read it as a signal word: ONTAP in a scenario usually means existing enterprise storage.

> ON THE EXAM
>
> Lustre and ONTAP are both FSx, and they answer completely different questions. Do not let the shared name blur them.

Dataset shape to storage and mode

WHAT YOU ARE GIVEN	WHAT TO REACH FOR
Small dataset, simple job	S3 with File mode. Do not overthink it
Large files, over 50 MB, read once	S3 with FastFile mode
Millions of small files, many repeat runs	FSx for Lustre linked to S3
Must shrink the training instance volume	S3 with Pipe mode
Data already written to a shared file system	Amazon EFS
Startup delay is the stated complaint	FastFile first. FSx only if it repeats

ON THE EXAM

Two words decide most of these: file size, and repetition. Read the scenario for both before comparing anything.

Constraints that break training jobs quietly

- **The dataset must be in the same Region** as the training job. Not negotiable.
- EFS and FSx both require the job to run **inside a VPC**. Forget it and the job cannot mount.
- The FSx subnet must map to the **same AZ** as the file system.
- The execution role needs read on the data and write on the output path.
- Cross-Region access is not a permissions problem. Copy or replicate the data first.

> ON THE EXAM
>
> When a scenario mentions two Regions and a failing training job, the Region mismatch is the answer, not IAM.

Ingestion problems, and what they usually are

SYMPTOM	USUAL CAUSE
Training starts slowly, then runs fine	File mode downloading the dataset. Change the mode
Throughput plateaus well below the instance limit	Too few S3 prefixes, or too many tiny objects
Job fails partway through, disk related	EBS volume too small for File mode
Job cannot reach the file system at all	No VPC configuration, or the wrong subnet AZ
Throttling on a DynamoDB extract	Scanning the live table instead of using export to S3
Works in one Region, fails in another	Dataset and training job in different Regions

ON THE EXAM

Read the symptom carefully. Slow start and slow throughout are different problems with different answers.

Storing text, images and audio for AI workloads

- **Text.** Small and cheap to store, but chunking and metadata decide retrieval quality later.
- **Images.** Large objects, often paired with a manifest holding labels and paths.
- **Audio.** Very large, usually converted to text or features before it is useful.
- All three land in S3. The difference is object size, and that drives the input mode.
- Keep the raw asset and the derived form. Reprocessing raw audio is expensive and slow.

> ON THE EXAM
>
> Multimodal data changes your file sizes, and file size is what decides File mode against FastFile mode.

Organising training data so it stays usable

ZONE	WHAT LIVES THERE
Raw	Exactly what arrived, never edited. Your only way back if processing was wrong
Curated	Cleaned, deduplicated, typed. Usually Parquet and partitioned
Feature-ready	The columns a model actually consumes. Often the Feature Store offline store
Datasets	Frozen, versioned training and validation splits tied to a model version

ON THE EXAM

Reproducing a model means reproducing its dataset. If you cannot point at the exact rows it trained on, you cannot.

Versioning a dataset, three ways

- **Immutable prefixes.** Write each snapshot to a new dated path. Simple and obvious.
- **S3 Versioning.** Keeps every version of a key. Watch the storage bill on large data.
- **Manifest files.** A list of the exact objects in a run. Cheap, precise, and portable.
- Overwriting training data in place destroys your ability to explain a past model.
- Record the dataset location in the model registry alongside the artefact.

> ON THE EXAM
>
> Auditors and incident reviews both ask the same question: what exactly did this model learn from?

The small-files problem

- Millions of tiny objects means millions of requests. Request overhead dominates.
- File mode download time balloons. FastFile does not help much either.
- **Fix at the source:** pack into larger shards, or a format like RecordIO-protobuf.
- Or use FSx for Lustre, which handles small files far better than S3 downloads.
- Aim for shards of tens to hundreds of MB rather than kilobytes.

> ON THE EXAM
>
> Two options can both be right here. Repacking the data is the durable fix. FSx is the fix when you cannot change the pipeline.

Datasets too big for a single machine

- **Shard the data** and give each training instance its own slice. This is the normal answer.
- SageMaker AI can distribute S3 keys across instances, or replicate the full set to each.
- Replicating a huge dataset to every instance multiplies both download time and disk need.
- Sharded input plus data parallel training is the standard large-dataset pattern.
- Streaming modes help here too, since no instance needs the whole dataset on disk.

> ON THE EXAM
>
> Distribution of data and distribution of computation are separate settings. Module 14 covers the compute half.

Extracting from operational databases

- **Amazon RDS.** Export a snapshot to S3 as Parquet, or read with a Glue job.
- **DynamoDB.** Export to S3 needs no capacity, and does not touch your read throughput.
- **Amazon Redshift.** UNLOAD to S3, or query through Athena federation.
- Do not point a training job at a production database. Extract first, always.
- Extracting also gives you a frozen snapshot, which is what reproducibility needs.

> ON THE EXAM
>
> Training directly against a live operational store is wrong twice over: it loads production, and the dataset is never reproducible.

Moving data in from outside

- **AWS DataSync.** Scheduled, managed transfer from on-premises file systems to S3 or EFS.
- **S3 Transfer Acceleration.** Uploads over long distances, routed via CloudFront edges.
- **Storage Gateway.** On-premises applications keep writing locally; data lands in S3.
- **AWS Snow family.** Physical transfer when the network genuinely cannot cope.
- Transfer Acceleration costs extra per GB. It is for distance, not for local uploads.

> ON THE EXAM
>
> Distance is the signal for Transfer Acceleration. Recurring on-premises sync is the signal for DataSync.

Block storage during training

WHAT	WHY IT MATTERS
EBS volume size	File mode needs room for the whole dataset. Pipe mode needs almost none
EBS Provisioned IOPS	For workloads where local disk read rate is the bottleneck
Instance store	Fast local NVMe on some families. Lost when the instance stops
Checkpoint location	Write checkpoints to S3, not local disk, or Spot interruption loses them

ON THE EXAM

Undersizing the EBS volume for File mode is a classic self-inflicted training failure.

Encryption and access, in one slide

- **At rest:** SSE-S3 by default, or SSE-KMS when you need your own key and an audit trail.
- **In transit:** TLS between the training job and S3, and between instances when configured.
- The **execution role** needs read on the data path and write on the output path. No more.
- Bucket policies and VPC endpoint policies can deny anything outside your network.
- Sensitive training data is a real exam theme. Modules 08 and 35 cover it properly.

> ON THE EXAM
>
> Mentioned here so the storage picture is complete. The detail, and the compliance angle, come later.

Pay for performance where the job is **actually waiting**, and nowhere else.

> A computer vision team stores 8 TB of training images in Amazon S3. The images average about 60 KB each, so there are well over a hundred million objects.
>
> They retrain daily on the same dataset. Each job currently spends around 45 minutes downloading before the first epoch begins, on eight GPU instances.

What should the team change?

A Keep File mode and increase the EBS volume size on the training instances

B Switch the training job to FastFile mode

C Create an FSx for Lustre file system linked to the S3 bucket and mount it

D Move the images to S3 Glacier Instant Retrieval to reduce storage cost

◆ PAUSE AND DECIDE

What should the team change?

A Keep File mode and increase the EBS volume size on the training instances

Volume size was never the constraint. The 45 minutes is request overhead across a hundred million small objects, not disk space.

B Switch the training job to FastFile mode

FastFile targets files over 50 MB. At 60 KB per object the per-request overhead dominates and the gain is small.

C Create an FSx for Lustre file system linked to the S3 bucket and mount it

Many small files plus daily repetition is exactly its case. Objects cache on first access, so later runs pay no download cost.

D Move the images to S3 Glacier Instant Retrieval to reduce storage cost

Answers a question nobody asked, and makes the real problem worse by adding retrieval cost to active training data.

THE SIGNAL

Two signals, both planted: tiny files, and daily repetition over the same dataset. Either alone is weaker. Together they point only at FSx.

A different team stores 4 TB of training video in Amazon S3. Each file is roughly 2 GB, so there are around two thousand objects.

They train once on each new batch and never revisit it. Jobs currently spend about 50 minutes downloading before training starts. The team has no VPC expertise and wants the smallest possible change.

What should this team do?

A Switch the training job to FastFile mode

B Create an FSx for Lustre file system linked to the S3 bucket and mount it

C Switch to Pipe mode and convert the videos to RecordIO-protobuf

D Copy the videos to Amazon EFS before each training run

◆ PAUSE AND DECIDE

What should this team do?

A Switch the training job to FastFile mode

Files are 2 GB, far above the 50 MB threshold. Training starts immediately, no file system, no VPC. One parameter changes.

B Create an FSx for Lustre file system linked to the S3 bucket and mount it

Correct in scenario 1, wrong here. Data is never revisited, so the caching never pays back, and it demands the VPC work they cannot do.

C Switch to Pipe mode and convert the videos to RecordIO-protobuf

Would remove the wait, but converting 4 TB of video is a large project. Nothing in the scenario asks to shrink the instance volume.

D Copy the videos to Amazon EFS before each training run

Replaces a download with a copy, so the wait moves rather than disappearing. It also needs the VPC setup they lack.

THE SIGNAL

Same complaint as scenario 1, opposite answer. Large files and no repetition flip it. That pair is worth remembering as a pair.

> A finance team provisioned a 1.2 TB FSx for Lustre file system eighteen months ago for a fraud model. It is linked to S3 and holds a 200 GB dataset.
>
> The model is now retrained once a month, and each run takes about twenty minutes. Finance has flagged the file system as one of the team's largest recurring line items.

What is the most cost-effective change?

A. Reduce the FSx for Lustre throughput per TB to the lowest available setting

B. Move the dataset to S3 Glacier Flexible Retrieval between monthly runs

C. Keep FSx but schedule it to be deleted and recreated around each monthly run

D. Delete the file system and train directly from S3 using File mode

◆ PAUSE AND DECIDE

What is the most cost-effective change?

A. Reduce the FSx for Lustre throughput per TB to the lowest available setting

Trims a bill that should be zero. You still pay the 1.2 TB minimum around the clock for twenty minutes of use a month.

B. Move the dataset to S3 Glacier Flexible Retrieval between monthly runs

Saves pennies on 200 GB and adds a restore of minutes to hours before every run. It also leaves the FSx bill untouched.

C. Keep FSx but schedule it to be deleted and recreated around each monthly run

Cuts the bill, but adds orchestration, a repopulation wait, and a fragile job to maintain, all to save minutes on a twenty-minute run.

D. Delete the file system and train directly from S3 using File mode

200 GB downloads quickly, and the job runs monthly. The performance FSx was buying is worth almost nothing at this cadence.

THE SIGNAL

FSx bills continuously; training here runs twenty minutes a month. The fix is not a cheaper file system, it is no file system.

A healthcare company is designing training infrastructure. Their compliance policy requires that all model artefacts be encrypted at rest with a customer-managed KMS key.

An engineer proposes writing training output to an S3 Express One Zone directory bucket, arguing that its low latency will speed up checkpoint writes during long training runs.

How should you respond?

A Approve it, and apply a bucket policy that requires the KMS key on every write

B Reject it. SSE-KMS is not supported for SageMaker AI output in directory buckets

C Approve it, but replicate the bucket to a second Availability Zone for durability

D Approve it, and encrypt the artefacts in the training script before writing them

◆ PAUSE AND DECIDE

How should you respond?

A. Approve it, and apply a bucket policy that requires the KMS key on every write

A policy cannot enable an unsupported feature. It would reject every write SageMaker AI attempts, breaking the job rather than securing it.

B. Reject it. SSE-KMS is not supported for SageMaker AI output in directory buckets

A documented service limitation, and the compliance requirement is non-negotiable. A general purpose bucket meets it directly.

C. Approve it, but replicate the bucket to a second Availability Zone for durability

Addresses a real weakness of One Zone storage, but not the one that blocks this. The encryption requirement is still unmet.

D. Approve it, and encrypt the artefacts in the training script before writing them

Shifts a managed control into hand-written code, losing key rotation, auditability and grants. Auditors treat these as different things.

THE SIGNAL

When a compliance requirement meets a service limitation, the limitation wins. Look for the option that changes the service, not the workaround.

What to take from this module

- **S3 is the default.** Leaving it needs a reason the scenario actually gives you.
- **File size and repetition** decide almost every input mode question.
- **FastFile above 50 MB.** FSx for small files read many times. File mode when it is simple.
- **FSx bills continuously.** One job a month cannot justify it.
- **Same Region, always.** And EFS or FSx means a VPC, every time.

> ON THE EXAM
>
> Module 03 covers the formats this data should be in, and the streaming path for data that never sits still.